Example 2.5 Find the greatest number that will divide 445 and 572 leaving remainders 4 and 5 respectively.

Solution Since the remainders are 4, 5 respectively the required number is the HCF of the number $445-4=441,\ 572-5=567$.

Hence, we will determine the HCF of 441 and 567. Using Euclid's Division Algorithm, we have,

$$567 = 441\times 1+126$$

$$441 = 126\times 3+63$$

$$126 = 63\times 2+0$$

Therefore, HCF of $441,567 = 63$ and so the required number is 63.



Activity 1

This activity helps you to find HCF of two positive numbers. We first observe the following instructions.

(i) Construct a rectangle whose length and breadth are the given numbers.

(ii) Try to fill the rectangle using small squares.

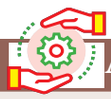
(iii) Try with 1×1 square; Try with 2×2 square; Try with 3×3 square and so on.

(iv) The side of the largest square that can fill the whole rectangle without any gap will be HCF of the given numbers.

(v) Find the HCF of (a) 12,20 (b) 16,24 (c) 11,9

Theorem 3

If a and b are two positive integers with $a > b$ then G.C.D of (a,b) = GCD of $(a-b,b)$.



Activity 2

This is another activity to determine HCF of two given positive integers.

(i) From the given numbers, subtract the smaller from the larger number.

(ii) From the remaining numbers, subtract smaller from the larger.

(iii) Repeat the subtraction process by subtracting smaller from the larger.

(iv) Stop the process, when the numbers become equal.

(v) The number representing equal numbers obtained in step (iv), will be the HCF of the given numbers.

Using this Activity, find the HCF of

(i) 90,15 (ii) 80,25 (iii) 40,16 (iv) 23,12 (v) 93,13

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